INCH-POUND

A-A-59384B
13 January 2014
SUPERSEDING
A-A-59384A
21 June 2006

COMMERCIAL ITEM DESCRIPTION

PRESS, "H" FRAME, HYDRAULIC

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) establishes the government acquisition requirements for floor mounted "H" frame presses which are used in maintenance and repair shops to press, straighten, assemble, and disassemble parts.

2. CLASSIFICATION. The "H" frame press shall be of the following types, styles, classes, and sizes.

2.1 Type. Type refers to the type of bed provided on the press. The type shall be as specified (see 7.2(b)) and is described herein as follows:

Type I - Non-roll bed
Type II - Roll bed (stationary frame)
Type III - Roll frame (stationary bed)

2.2 Style. Style refers to the style of cylinder used on the press. The style shall be as specified (see 7.2(c)) and is described herein as follows:

Style 1 - Single stage cylinder
Style 2 - Dual stage cylinder

2.3 Class. Class refers to the method of operation. The class shall be as specified (see 7.2(d)) and is described herein as follows:

Class A - Hand operated
Class B - Electric operated
Class C - Air operated

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or DLA Aviation - VEB, 8000 Jefferson Davis Hwy, Richmond VA, 23297-5616. Since contact information can change, you may want to verify the currency of this address information using ASSIST database at https://assist.dla.mil.

FSC 3442

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2.4 Size. Size refers to the pressing capacity of the press. The size shall be as specified (see 7.2(e)) and is described herein as follows:

010 - 10 ton capacity	040 - 40 ton capacity	080 - 80 ton capacity
017 - 17 ton capacity	050 - 50 ton capacity	100 - 100 ton capacity
020 - 20 ton capacity	055 - 55 ton capacity	150 - 150 ton capacity
025 - 25 ton capacity	060 - 60 ton capacity	200 - 200 ton capacity
030 - 30 ton capacity	075 - 75 ton capacity	300 - 300 ton capacity

3. SALIENT CHARACTERISTICS

3.1 Design. The press shall be new and of the "H" frame design. The press shall be designed for floor mounting and shall include all components, controls, parts, and features to meet the performance requirements specified herein. The press shall comply with the applicable requirements of ANSI B11.2, for the press ordered.

3.2 Controls. The press shall be equipped with all the controls required for the complete control of pressing operations. The controls shall provide for rapid advancement of the ram to and retraction from the work piece. All operating controls shall be located convenient to the operator at his normal workstation.

3.3 Frame. The frame shall be fabricated of structural steel sections consisting of a bolster, work bed, side rails, and required support structures. All components shall be fastened together with bolts or by welding or a combination of bolting and welding. The bolster shall support the hydraulic head assembly and shall provide the strength to maintain parallelism of the side members. The press frame shall have the strength to withstand not less than 150 percent of the maximum rated capacity of the press. The frame shall be supported by structures of the length and strength to provide stability to the press when loaded to maximum pressing capacity. The maximum and minimum ram nose to table dimensions shall be as specified (see 7.2(f)).

3.3.1 Siderails and support pins. The siderails shall have a series of equally spaced holes to accommodate pins to support the work bed or bolster. No less than two steel pins shaped for easy insertion and removal from the holes shall be provided. The shear strength of the pins shall be not less than 150 percent of the maximum rated capacity of the press. The pin hole spacing in all members shall be positioned to ensure that all pins support equal loads when pressing force is centered between the bolster and the work bed. Minimum width between the side rails shall be as specified (see 7.2(g)).

3.3.2 Type I: Non-roll bed. Type I non roll bed presses shall have beds with matching holes to align with the holes in the siderails for accommodating the support pins. The minimum width between bed channels shall be as specified (see 7.2(h)). The bed shall have the strength to withstand the working forces when the press is operated at 150 percent of rated capacity. A hydraulic or mechanical means for raising and lowering the work bed shall be provided.

3.3.3 Type II: Roll bed. Type II roll bed presses shall have beds mounted in tracks that allow the bed to roll out from under the bolster. The roll bed travel distance shall be such that when the

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bed is in the loading position, all bed edges shall be outside of a perpendicular line from the side of the bolster structure closest to the extended bed. The roll bed shall be provided with a locking mechanism to secure the bed in position. The bed shall have the strength to withstand the working forces when the press is operated at 150 percent of rated capacity. The minimum width between bed channels shall be as specified (see 7.2(h)). When specified (see 7.2(i)), a bed with adjustable channel width shall be supplied. A hydraulic or mechanical means for raising and lowering the bolster shall be provided.

3.3.4 Type III: Roll frame. Type III roll frame presses shall be mounted on rollers that roll along the table axis in a track mounted on the side of the press table. The frame shall be provided with a locking mechanism to lock the frame in position. The press shall have the strength to withstand the working forces when operated at 150 percent of rated capacity. A hydraulic or mechanical means for raising and lowering the bolster shall be provided.

3.4 Hydraulic system. The hydraulic system shall conform to the requirements specified herein and International Organization for Standardization (ISO) ISO-4413. The safe operating pressure rating of all hydraulic components shall be 110 percent of the maximum rated capacity of the press. The hydraulic system shall consist of a hydraulic head assembly, fluid reservoir, pressure gauge, and the necessary fittings and hoses to ensure a complete hydraulic system. The hydraulic system shall be either self-bleeding or have provisions for bleeding air from the system after the system is filled.

3.4.1 Hydraulic head assembly. The hydraulic head assembly shall support the cylinder and ram assembly. Unless otherwise specified (see 7.2(j)), the hydraulic head assembly shall be permanently fixed to the center of the bolster. Moveable hydraulic head assemblies shall use roller bearings and shall be adjustable laterally along the bolster. A means shall be provided to lock the moveable hydraulic head assembly at any position within its range of travel. The moveable hydraulic head assembly shall be capable of performing full rated pressing operations at any position within its range of travel.

3.4.2 Cylinder and ram assembly. The cylinder and ram assembly shall be furnished with either a dual stage or single stage cylinder as specified (see 7.2(k)). The minimum ram stroke length shall be as specified (see 7.2(l)). Provisions to prevent extension of the ram beyond its normal end-of-stroke position shall be provided. Hydraulic ram assemblies shall meet the requirements of ASME B30.1. The ram axis shall be perpendicular to the worktable within an accuracy of 0.120 inch total indicator reading per foot of travel.

3.4.3 Fluid reservoir. The hydraulic fluid reservoir shall be capable of holding not less than 1-1/4 times the volume of fluid required to fill the hydraulic system and force the ram to the bottom of its stroke. If the reservoir is pressurized during any part of the operating cycle, the reservoir shall have a safe operating pressure rating of not less than twice the pressure that can be imposed on the interior of the reservoir.

3.4.4 Pressure gage. A pressure gage calibrated to read in tons of ram force shall be furnished. The upper range of the gage shall be not less than 150 percent of the maximum system pressure. The pressure gage shall comply with the requirements of ASME B40.100 Grade B.

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3.4.5 Pressure relief valve. The hydraulic system shall have a pressure relief valve or similar device to prevent loading the press beyond 110 percent of its rated capacity.

3.5 Work support devices. The press shall be furnished with two flat work support devices and two "V" type support devices. The devices shall be movable along the entire length of the work bed. These devices shall have a means to prevent them from sliding off the work bed. A single pair of invertible support devices suitable for supporting flat work pieces on one side and round work pieces on the other side may be furnished in lieu of the two pairs. Both types of work support devices shall withstand repeated pressing forces. One flat ram nose and one "V" ram nose shall be supplied.

3.6 Class A: hand operated hydraulic press. Hand operated presses shall conform to the following requirements.

3.6.1 Hydraulic pump. The hand operated hydraulic pump shall be capable of producing 10,000 pound-force per square inch (psi) when operated on shop air pressure of 80 to 100 psi. Maximum handle effort shall be less than 100 pounds.

3.7 Class B: Electric Motor Operated Hydraulic Press. Electric motor operated hydraulic presses shall conform to the following requirements.

3.7.1 Power requirements. Electric operated presses shall be wired to conform to National Fire Protection Association (NFPA) NFPA-79. Each press shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 7.2(m)), each press shall be designed to operate on 3-phase 60-Hertz (Hz) dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components.

3.7.2 Motors. The manufacturers standard motor provided to the commercial market shall be supplied.

3.8 Class C: Air Operated Hydraulic Press. Air operated hydraulic presses shall conform to the following requirements.

3.8.1 Air Driven Hydraulic Pump. The power and performance characteristics of the air driven hydraulic pump and other components of the pneumatic system shall be sufficient to operate the press at 110 percent of maximum rated press capacity as specified herein when operating on a shop air pressure of 80 to 100 psi. When specified (see 7.2(n)), an air pressure regulator, pressure gauge, air filter, water separator, lubricator, and the necessary operating valves shall be provided. The pneumatic power system shall be mounted as an integral part of the hydraulic head assembly or as a separate unit mounted on the press frame. The pneumatic power system shall comply with the requirements specified herein and the requirements of ISO-4414.

3.9 Standard equipment. All standard equipment normally provided with the manufacturer's commercial press shall be furnished.

3.10 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described by the procuring activity (see 7.2(o)).

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3.11 Physical size limitations. When specified (see 7.2(p)), the physical size of the press shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the press will fit into its future operating location.

3.12 Marking on instruments, control panels, charts, and plates. All words on the press, including all instruments, control panels, charts, plates, and related attachments shall be in the English language. Characters shall be marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.13 Data plates. All standard data plates normally provided with the manufacturer's standard commercial press shall be furnished. Any additional data plates shall be as specified and fully described by the procuring activity (see 7.2(q)).

3.13.1 Nameplate. Unless otherwise specified (see 7.2(r)), a nameplate shall be securely attached to each press. The nameplate shall contain the information listed below.

a. Nomenclature.
b. Manufacturer name.
c. Manufacturer model.
d. Manufacturer serial number.
e. Contract number or order number.
f. National stock number.
g. Date of manufacture.

3.14 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in FED-STD-376, and all other requirements of this commercial item description are met.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance the Federal Acquisition Regulation (FAR) FAR-23.403.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

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6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.2(s)).

7. NOTES

7.1 Sources of referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Electronic copies of may be obtained from https://assist.dla.mil/.

Occupational Safety and Health Administration (OSHA)

OSHA 29 CFR Part 1910	Occupational Safety and Health Standards for General Industry

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210. Electronic copies of OSHA standards may be obtained from https://www.osha.gov/.

Federal Acquisition Regulation (FAR)

FAR-23.403	Use of Recovered Materials and Biobased Products

The FAR may be obtained from the US Government Printing Office 732 North Capitol Street N.W., Washington, DC 20401. Electronic copies of FAR standards may be obtained from http://www.acquisition.gov/far/.

Federal Standard (FED-STD)

FED-STD-376	Preferred Metric Units for General Use by the Federal Government

Applications for copies should be sent to the General Services Administration, Federal Supply Service, Environmental and Engineering Policy Division, Washington, DC 20406. Electronic copies of Federal Standards may be obtained from https://www.assist.dla.mil/.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.2	Hydraulic Power Presses - Safety Requirements for Construction, Care, and Use

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Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036. Electronic copies of ANSI standards may be obtained from http://www.ansi.org/.

ASME

ASME-B30.1	Jacks, Industrial Rollers, Air Casters, and Hydraulic Gantries
ASME B40.100	Pressure Gauges and Gauge Attachments, Pressure

Applications for copies should be sent to the ASME , Three Park Ave., New York, NY 10016-5990. Electronic copies of ASME standards may be obtained from http://www.asme.org/.

National Electrical Manufacturers Association (NEMA)

ANSI/NEMA MG-1	Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209. Electronic copies of NEMA standards may be obtained from http://www.nema.org/.

National Fire Protection Association (NFPA)

ANSI/NFPA 79	Electrical Standard for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101. Electronic copies of NFPA standards may be obtained from http://www.nfpa.org/.

International Organization for Standardization (ISO)

ISO-4413	Hydraulic Fluid Power – General rules and safety requirements for systems and their components
ISO-4414	Pneumatic Fluid Power – General rules and safety requirements for systems and their components

Applications for copies should be sent to the, Customer Service Division, 545 Washington Boulevard, Jersey City, NJ 07310-1686. Electronic copies of ISO standards may be obtained from http://www.iso.org/.

7.2 Ordering data. The acquisition order should specify the following:

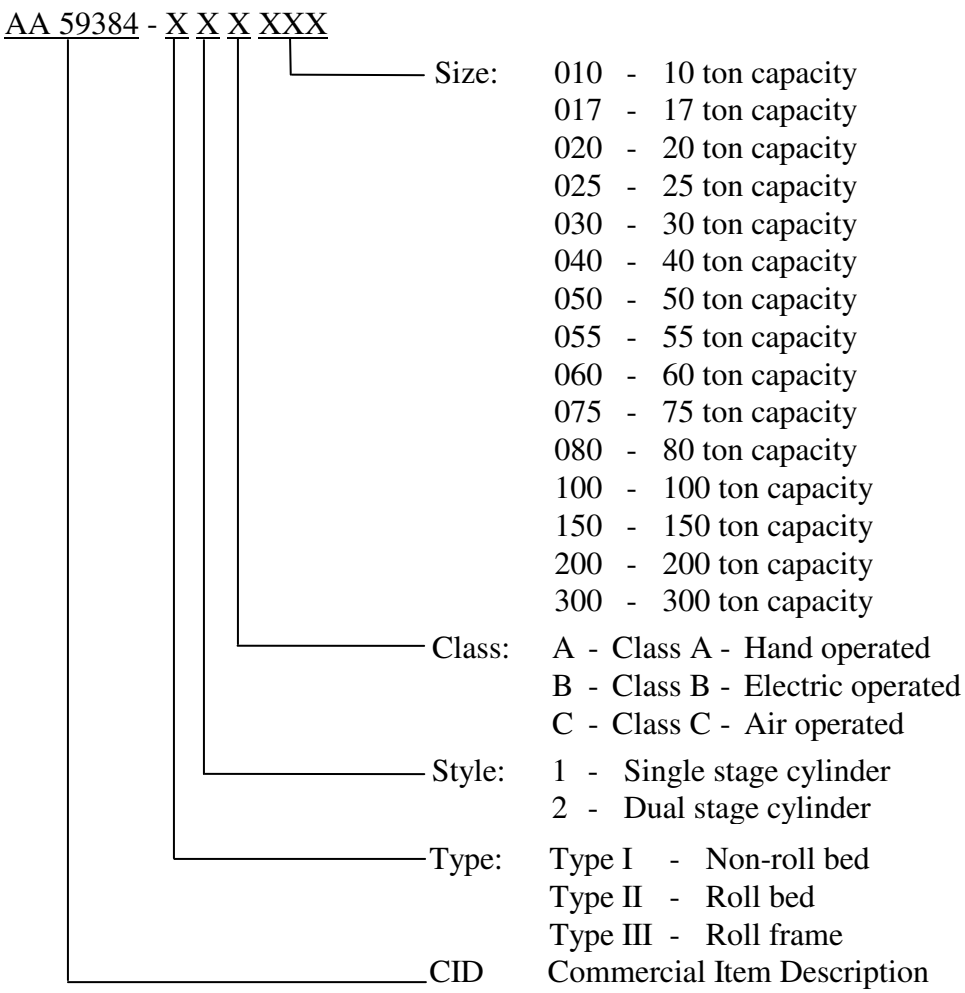
a. CID document number, revision, and CID PIN.
b. Type (see 2.1).
c. Style (see 2.2).
d. Class (see 2.3).

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e. Size (see 2.4).
f. Maximum and minimum ram nose to table dimensions (see 3.3).
g. Minimum width between side rails (see 3.3.1).
h. Minimum width between bed channels (see 3.3.2 and 3.3.3).
i. Adjustable width bed channel, if required (see 3.3.3).
j. Movable head assembly, if required (see 3.4.1).
k. Dual stage or single stage cylinder (see 3.4.2).
l. Minimum ram stroke length (see 3.4.2).
m. Electric power requirements, if different (see 3.7.1).
n. Air pressure regulator, pressure gauge, air filter, water separator, lubricator, and valves, if required (see 3.8.1).
o. Optional equipment and description, if required (see 3.10).
p. Physical size limitations, if required (see 3.11).
q. Additional data plate and description, if required (see 3.13).
r. Nameplate requirements, if different (see 3.13.1).
s. Preservation, packing, and marking (see 6.1).

7.3 Part Identification Number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

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7.4 Subject term (key word) listing.

air operated
double action cylinder
electric operated
hand operated
roll bed
roll frame
siderails
single action cylinder
stationary bed
stationary frame

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MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA - GS6

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